

# CS 3450 — Operating Systems

Department of Computer Science, Ridgeline State University — Fall 2026

**Instructor:** Dr. Maya Ellison (ellison@ridgeline.edu)

**Lecture:** Mon/Wed/Fri 10:00–10:50 AM, Hobbs Hall 214

**Lab:** Thursdays 2:00–3:50 PM, Hobbs Hall 120

**Office hours:** Tuesday & Thursday 11:00 AM–12:00 PM, Hobbs Hall 318

**Term:** Classes run Monday, August 24 through Friday, December 11, 2026.

## Course Description

An introduction to the design and implementation of modern operating systems: processes and threads, CPU scheduling, synchronization, deadlock, memory management, file systems, I/O, and virtualization. Prerequisite: CS 2420.

## Grading

| Component                         | Weight |
|-----------------------------------|--------|
| Homework (5 assignments, 6% each) | 30%    |
| Quizzes (3 quizzes, 4% each)      | 12%    |
| Midterm Exam                      | 20%    |
| Final Project                     | 13%    |
| Final Exam                        | 25%    |

All homework is submitted on Gradescope and is due at 11:59 PM on the date listed.

## Grade Scale

| A      | A–    | B+    | B     | B–    | C+    | C     | D     | F   |
|--------|-------|-------|-------|-------|-------|-------|-------|-----|
| 93–100 | 90–92 | 87–89 | 83–86 | 80–82 | 77–79 | 73–76 | 60–72 | <60 |

## Weekly Schedule

| Week | Dates            | Topics                                    | Due / In Class            |
|------|------------------|-------------------------------------------|---------------------------|
| 1    | Aug 24–28        | Introduction; processes                   | —                         |
| 2    | Aug 31–<br>Sep 4 | Threads and concurrency                   | Homework 1 due Fri, Sep 4 |
| 3    | Sep 7–11         | CPU scheduling (no class Mon — Labor Day) | —                         |

|    |                  |                                                  |                                                                         |
|----|------------------|--------------------------------------------------|-------------------------------------------------------------------------|
| 4  | Sep 14–18        | Synchronization                                  | Quiz 1 in class Wed, Sep 16                                             |
| 5  | Sep 21–25        | Deadlock                                         | Homework 2 due Fri, Sep 25                                              |
| 6  | Sep 28–<br>Oct 2 | Memory management                                | —                                                                       |
| 7  | Oct 5–9          | Virtual memory                                   | Homework 3 due Fri, Oct 9                                               |
| 8  | Oct 12–16        | Review; <b>Midterm Exam in class Wed, Oct 14</b> | Midterm Exam Wed, Oct 14                                                |
| 9  | Oct 19–23        | File systems                                     | —                                                                       |
| 10 | Oct 26–30        | File system implementation                       | Quiz 2 in class Wed, Oct 28                                             |
| 11 | Nov 2–6          | I/O systems                                      | Homework 4 due Fri, Nov 6                                               |
| 12 | Nov 9–13         | Virtualization                                   | —                                                                       |
| 13 | Nov 16–20        | Distributed systems                              | Quiz 3 in class Wed, Nov 18                                             |
| 14 | Nov 23–27        | Thanksgiving break — no class                    | —                                                                       |
| 15 | Nov 30–<br>Dec 4 | Security                                         | Homework 5 due Fri, Dec 4                                               |
| 16 | Dec 7–11         | Wrap-up and review                               | Final Project due Mon, Dec 7; <b>Final Exam Fri, Dec 11 at 10:00 AM</b> |

## Policies

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Late homework loses 10% per day, up to three days. Quizzes and exams cannot be made up without a documented excuse. Collaboration is encouraged on concepts, but all submitted code must be your own.